CSCI 325 - Data Structures & Algorithms

Lab: Sorting and Searching Practice

Purpose

Practice tracing and implementing the sorting and searching algorithms covered in class, and connect binary search with its requirement that the data be sorted.

Activity 1 : Algorithm Detectives

A. Trace binary search for target 42 in the sorted data:

{5, 11, 18, 26, 42, 57, 63}

Step	low	high	mid	Value checked / decision

Final index of 42: _____

B. Sorting Algorithms' Tracing:

Consider the vector **{9, 4, 6, 2, 8}** and perform the following operations:

Algorithm	Operation	Result
1. Insertion Sort	After iteration 1	_____
	After iteration 2	_____
2. Bubble Sort	After pass 1	_____
	After pass 2	_____
3. Merge Sort	After first division	_____
	After second division	_____
	Show the first merge	_____

Activity 2 : Coding Practice

Start with the following vector:

vector<int> values = {14, 5, 9, 2, 11, 7};

1. Write a bubble sort that sorts values in ascending order.
2. Display the vector before and after sorting.
3. Write/use binary search to search for target 11. Report the resulting index or boolean true/false.
4. Run the search again with one target that is not present in the vector.
5. Briefly explain why binary search should be performed after sorting in this activity.